# Domain Generalization via Aggregation and Separation for Audio Deepfake Detection

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***Abstract*— In this paper, we propose an Aggregation and Separation Domain Generalization (ASDG) method for Audio DeepFake Detection (ADD). Fake speech generated from different methods exhibits varied amplitude and frequency distributions rather than genuine speech. In addition, the spoofing attacks in training sets may not keep pace with the evolving diversity of real-world deepfake distributions. In light of this, we attempt to learn an ideal feature space that can aggregate real speech and separate fake speech to achieve better generalizability in the detection of unseen target domains. Specifically, we first propose a feature generator based on Lightweight Convolutional Neural Networks (LCNN), which is employed for generating a feature space and categorizing the feature into real and fake. Meanwhile, single-side domain adversarial learning is leveraged to make only the real speech from different domains indistinguishable, which enables the distribution of real speech to be aggregated in the feature space. Furthermore, a triplet loss is adopted to separate the distribution of fake speech while aggregating the distribution of real speech. Finally, in order to test the generalizability of the model, we train it with three different English datasets and evaluate in harsh conditions: cross-language and noisy datasets. The extensive experiments show that ASDG outperforms the baseline models in cross-domain tasks and decreases Equal Error Rate (EER) by up to 39.24% when compared to that of RawNet2. It is proved that the proposed Aggregation and Separation Domain Generalization method can be an effective strategy to improve the model generalizability.**

***Index Terms*— Audio deepfake detection, domain generalization, feature generator, triplet loss.**

## I. Introduction

WITH the progression of deep learning in recent years, Text-to-Speech (TTS) [1] and Voice Conversion (VC) [2] have been rapidly developed. The adversarial technology, Audio DeepFake Detection (ADD), faces serious challenges [3], [4]:

- The existing theoretical studies on the differences between real and fake speech are inadequate. It is difficult to explain the difference between real and fake at feature or signal level.
- Current ADD models suffer from poor generalization ability due to the lack of large-scale and diverse datasets.
- Deep learning methods usually perform better than traditional methods in ADD task but lack of interpretability.

In this paper, we focus on improving the generalizability of a model while enhancing the interpretability of the hidden states in the model.

Early ADD approaches showed promising results in intra-domain database testing [5], [6], [7], [8], [9], [10]. However, in real scenario, these approaches may considerably struggle in cross-domain situations which result from the channel effects of the speech [11]. The discrimination ability of an ADD model is affected by different domains such as different channels, codecs, compression formats, recording devices, and so on [12].

To improve the generalizability of ADD model in cross-domain dataset, Data Augmentation (DA) techniques are proposed to increase the diversity of data distribution. Recently, an increasing number of studies have concentrated on DA in ADD tasks [13], [14]. Although the above studies can improve the discriminatory ability of the model in cross-domain situations, there are too many unseen spoofing attacks that are not detected in the training data in realistic scenarios. It is inefficient to add new bonafide or spoofed audio in training data to achieve a better result. In a cross-domain ADD task, there is a significant risk. Specifically, the model may produce inaccurate determinations since the test utterances are unseen in the training dataset. Similarly, new algorithms of TTS and VC are proposed rapidly which led to more and more attack types. Hence, only DA method alone will never be sufficient. We need to learn an ideal feature space from the limited but known datasets so that the unseen target domain can be properly mapped in this space.

On the other hand, Domain Generalization (DG) methods have been proposed to generalize to unseen target domains well. Domain-Invariant Representation Learning (DIRL) is a prominent technique in DG. The goal of DIRL methods is to align the distributions across several source domains in

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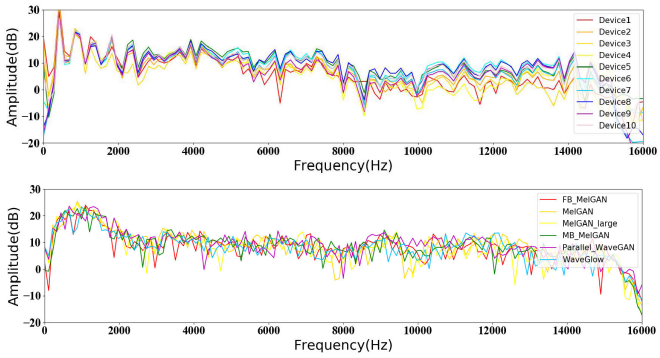


Fig. 1. Top: magnitude spectra of real audio which simulated from ten different devices. Bottom: Fake audio spectra with same linguistic information generated from six spoofing strategies.

order to develop a generalized feature space. In the field of face anti-spoofing and face forgery detection, the identification results are not generalizable in cross-domain experiments which result from the variations of environmental illumination, camera specifications, different spoofing attack types and so on. Thus, to improve the model generalizability, DIRL strategies are widely applied in face anti-spoofing and face forgery detection in recent years [15], [16], [17], [18], [19], [20]. Likewise, in the ADD task, poor results often arise owing to variations of environmental noise, recording equipment specifications and spoofing attack methods. Therefore, DIRL methods may also be useful to ADD in real scenarios. We can learn a compact feature space by analyzing the differences between real and fake speech. For instance, as shown in Fig. 1, we compare the real speech and fake speech with the same content. The real speech is selected from WaveFake real subsets [21] and an open-source simulator [22] is utilized to simulate ten real-world scenarios, while fake utterances are generated using six different spoofing techniques. Real speech exhibits minimal variations in its amplitude-frequency curve across 10 different simulated devices, with only slight differences in amplitude. In contrast, fake speech shows significant variations in the amplitude-frequency curve across different methods of falsification. As techniques improve and more spoofing methods emerge, an increasing number of amplitude-frequency curves can be formed for fake speech. Thus, in the ideal feature space, the proportion of the fake speech distribution should be much larger than that of real speech, with real speech represented as a single cluster, regardless of domain, and fake speech widely scattered over a larger area.

Based on the aforementioned viewpoints, we propose an Aggregation and Separation Domain Generalization (ASDG) method for ADD task. We aim to aggregate the real speech distribution through a single side adversarial domain discriminator and separate the fake speech by triplet mining. The input of the domain discriminator includes only real datasets rather than fake ones to make the features of real speech from different domains undistinguishable. Simultaneously, the triplet mining technique can effectively discrete the spoofing features and aggregate the real features. The experimental results with a cross-dataset show that our proposed model can achieve the lowest EER of 5.70% in clean condition and outperforms baseline models.

The main contributions of this work are as follows:

- To the best of our knowledge, we are the first to introduce the DIRL strategy for the ADD task to improve the model generalizability.
- We propose an ideal feature space recognition for real and fake speech feature distribution. An aggregation and separation domain generalization network with the modified triplet loss and single domain adversarial loss is designed to enhance the interpretability of the forensic network.
- Numerous experiments are designed in harsh conditions, i.e., cross-language and noisy datasets. The experiments show that ASDG outperforms the baseline models in cross-domains and decreases the EER by up to 39.24%, verifying the generalizability of our model for unknown target domains.

## II. Related Work

This section presents a brief introduction to the development of audio deepfake approaches, audio deepfake detection methods and typical domain generalization methods.

### A. Audio Deepfake Approaches

With the increasing modelling power of neural networks, the performance of TTS and VC generated speech is getting closer to that of a human recording.

WaveNet, an early audio deepfake approach with a vocoder in TTS [23], generates waveforms from linguistic features by using dilated causal convolution. To generate speech directly from text, Tacotron and Tacotron2 were proposed. Tacotron [24] is an innovative end-to-end TTS system that leverages an encoder–attention–decoder framework and uses the Griffin–Lim algorithm to generate waveforms. In Tacotron2 [25], the audio quality is greatly improved by using WaveNet as a vocoder. However, the autoregressive generation strategy in Tacotron results in its slow inference speed. FastSpeech [26], which was proposed to solve this issue, adopts a transformer network to generate a mel spectrogram in parallel. To achieve better quality and controllable speech synthesis, FastSpeech2 [27] was proposed. Recently, Kim et al. introduced a novel method named VITS [28], which enhances the expressive capacity of generative modelling by incorporating variational inference with normalizing flows and employing an adversarial training process. Furthermore, Tan et al. proposed Natural Speech [29]. No statistically significant difference from human recordings was realized for the first time using Natural Speech. For the vocoder in TTS, GANs have been widely used. GANs consist of a generator for data generation and a discriminator to determine the authenticity of the generated data. Representative GANs include MelGAN [30], Parallel WaveGAN [31], Multiband MelGAN [32] and HiFi-GAN [33]. In recent years, several researchers have explored the application of denoising diffusion probabilistic models (DDPMs) [34] in the field of vocoders. Notable examples include DiffWave [35] and

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WaveGrad [36], which leverage DDPM techniques to enhance the performance of vocoders.

Compared to TTS, VC models have been rapidly developed in terms of both accuracy and comprehensiveness. These approaches can be broadly categorized into parallel VC [37], [38] and nonparallel VC [39], [40] methods. Parallel VC methods rely on paired data where the source and target utterances share the same content and learn a direct mapping from the source to the target voice using various techniques. On the other hand, nonparallel VC techniques operate without paired training data. They aim to learn the mappings between different voices by exploiting unpaired datasets. Approaches such as StarGAN-VC [39], CycleGAN-VC [40] and phonetic posteriorgram (PPG)-based approaches [41], [42] fall into this category. In recent years, one-shot VC has received increasing attention due to its well-generalized performance for unseen target speakers. For instance, AutoVC [43] was the first to perform zero-shot VC, which utilizes a vanilla autoencoder with a carefully designed bottleneck. Wu and Lee [44] proposed a U-Net architecture within an autoencoder-based VC system. Chen et al. [45] used activation guidance and adaptive instance normalization based on a single encoder and decoder. Overall, the current TTS and VC deepfake technologies are highly developed and can achieve high-quality speech forgeries.

### *B. Audio Deepfake Detection*

With the fast propagation and high-quality of deepfake audio, the audio deepfake detection has attracted increasing attention from endeavors. Since 2015, worldwide ASVspoof challenges have been held four times. And then the first Audio Deep Synthesis Detection Challenge was also successfully held. In ASVspoof challenges, an LA (Logical Access Scenario) subset is concentrated on deepfake speech which converted by TTS and VC. ADD task is starting to evolve rapidly around the ASVspoof2019 LA dataset. The models proposed to access ADD task can be categorized in two classes: methods focus on the extracting speech features and methods based on the deep learning classifier.

For the first category, the researchers have attempted to use different handcrafted speech features in ADD which conclude Mel-filter frequency Cepstral coefficients (MFCC), Linear filter frequency Cepstral coefficients (LFCC), Constant Q Cepstral Coefficients (CQCC) [46], and other features from deep learning network [47], [48]. More recently, researchers have found that utilizing large-scale self-supervised models, such as Wav2Vec2 (W2V2) [49], as the front-end, which leads to excellent results in the ADD domain [50]. Especially in ADD2022 challenge [51], W2V2 exhibits robust capabilities in low-quality environments [52] or partially spoofed ADD condition [53].

For the second category, several classifier utilizing deep neural network models have achieved promising results with ASVSpoof 2019 LA dataset. Lavrentyeva et al. [54] utilize Lightweight Convolutional Neural Networks (LCNN) as backbone. Zhang et al. [55] concentrated on capturing the real speech class distribution. They set a tight classification boundary using a one class loss function and use ResNet18 as backbone. The one of state-of-art backbone in speaker recognition tasks like ECAPA-TDNN has been also employed in ADD task [10]. Tak et al. [6] attempted to use raw audio rather than any handcrafted feature as the input of network and utilize RawNet2 as classifier backbone. RawGAT-ST [8] and AASIST [56] are improved version based on RawNet2, that introduce a novel spectral-temporal graph attention network and a heterogeneous stacking graph attention network, respectively.

Although deep neural networks are particularly proficient at dealing with classification issues, it can be challenging to provide a convincing explanation for why it works so well. Furthermore, most models suffer from poor performance in cross-domain classification scenario, owing to training and test domain mismatch. To deal with these issues, researchers have attempted to use Data Augmentation (DA) strategy to expand the training distribution in order to closely approximate the target domain.

From the ASVspoof 2021 challenge [57], the organizers of the competition aimed to improve the generalizability of the ADD model. Therefore, speed perturbation [58], SpecAugment [59] and codec augmentation [60] were used in the evaluation data to help improve the cross-domain identification ability. Linear and nonlinear convolutive and additive noise were used to augment the datasets in RawBoost [13]. Twelve different channels were simulated from a single dataset to improve the model performance with a cross-dataset [11] condition. Cohen et al. [12] proposed two forms of DA, compression and channel augmentation, and introduced a new type of online DA, SpecAverage. In [14], a combination of three deepfake and spoofing databases named the Attack Agnostic Dataset was used to increase the training stability and general diversity of the combined datasets. Although most of the aforementioned studies have employed domain adaptation approaches, there has been no investigation into the interrelationships between different training domains to enhance the generalizability of these approaches, such as the DIRL method.

### *C. Domain-Invariant Representation Learning*

DA can increase the diversity of training data to enhance the model generalizability. However, in real scenario, speech signal is very complex due to the large variations in frequency and amplitude. The use of only a few datasets for expansion is not sufficient to achieve good performance in the presence of an unseen target domain. We expect to learn an ideal feature space from the limited datasets which is the reason for utilizing DIRL. In DIRL, it is assumed that there exists a generalized feature space that can easily discriminate the seen source domain and the unseen target domain. We can learn the feature space from the source domain and generalize to unseen target domains well. DIRL is a popular strategy for Face Anti-Spoofing (FAS) and Face Forgery Detection (FFD) in recent years. Li et al. [61] use Maximum Mean Discrepancy strategy to learn generalized features for FAS. Shao et al. [62] propose a multi-adversarial domain generalization framework to learn a generalized feature space by source domains. Wang et al. [63] propose to disentangle generalized FAS features from subject discriminative and domain-dependent features. According

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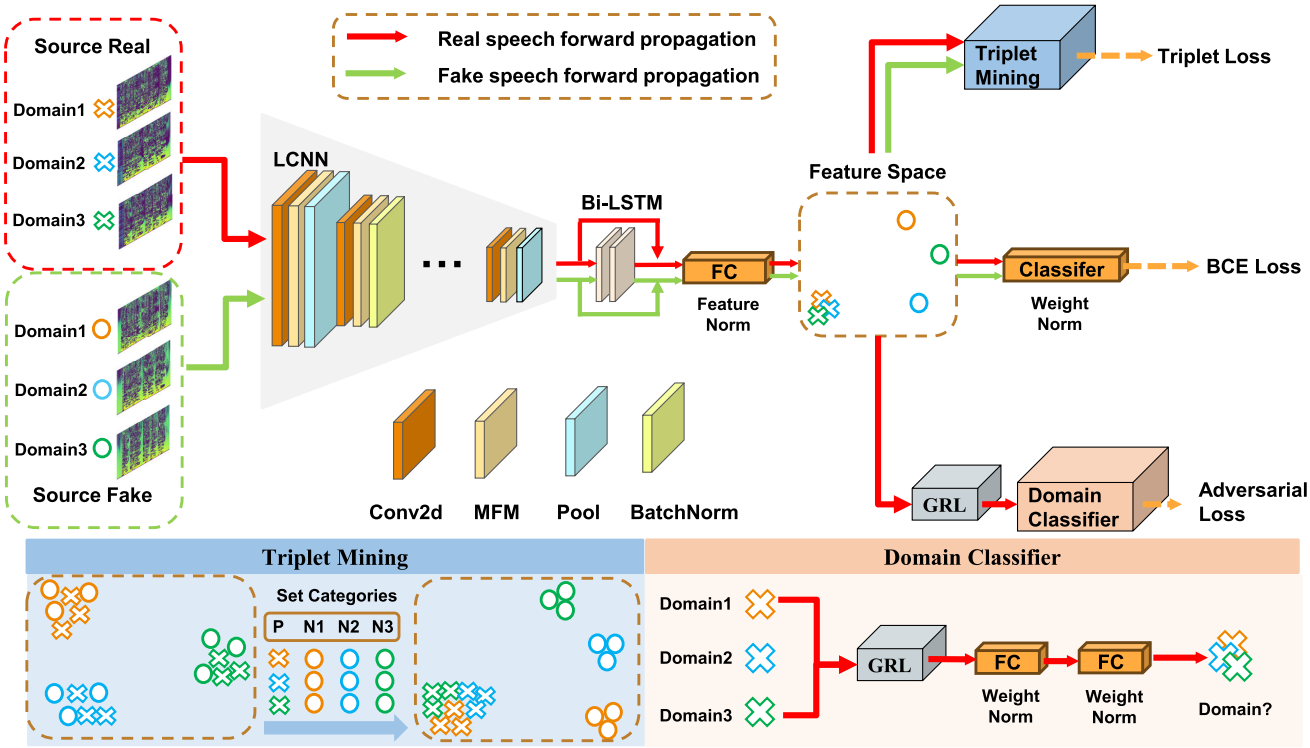


Fig. 2. The whole architecture of our proposed approach. The red and green lines represent the forward propagation routes of real and fake speech respectively. The classifier composed of FC will compress the features to 1 dimension for the binary cross entropy loss, meanwhile, the triplet mining is implemented to optimize the Feature space. The real speech feature will be send to the domain classifier to identify the domain and use gradient reverse layer (GRL) to make the real feature undistinguishable.

to evidence that instance normalization can remove domain discrepancy, Liu et al. [64] utilized aggregate batch and instance normalizations for generalized representation. These successful applications have convinced us of the effectiveness of the DIRL approach to improve the model generalizability in cross-domain scenario.

## III. Proposed Method

### A. Overview

Since there are more and more new spoofing attacks generated with TTS and VC algorithms and unseen encoding, transmission, or other conditions. The occurrence of logic access spoofing attacks in the training set may not keep pace with the increasing diversity of the spoofing attack distributions in real-world scenarios. Thus, we should concentrate on the real speech by aggregating the real speech distribution while separating the fake one. In this section, we introduce our Aggregation and Separation Domain Generalization network, which is shown in Fig. 2. Firstly, we present the Feature generator in our network to extract the LFCC into a feature space. In order to enhance the diversity of the forensic dataset, the input LFCC are extracted from three different datasets. Secondly, a single-side adversarial domain classifier is proposed to identify which domain the feature belongs to. In this way, we can aggregate the real speech regardless of the speech domain. Then, to separate fake features while aggregating real features, a modified triplet mining method is added. Finally, the overall loss is integrated to optimize the network.

### B. Feature Generator

In order to obtain an ideal feature space from a limited dataset, a well-performing feature generator is essential. In feature generator, we use the LCNN [54] and Bidirectional LSTM (Bi-LSTM) [65] to extract the feature space from the source domain speech. The detail of our LCNN based structure is shown in Table. I. We concatenate the real and fake LFCC as the input of LCNN which from three different domains. To capture the speech information on time series, two Bi-LSTM layers with skip connection are added after LCNN. Then, a Fully Connected (FC) layer is used to compress the feature to 512. Finally, it should be noted that the feature normalization is used for the output of the feature generator. On one hand, under both intra-database and cross-database scenarios, there are significant discrepancies between the feature norms of various speech, which makes the process of feature learning difficult.

On the other hand, normalization can ensure that loss values with the same magnitude are obtained, which will help the multi-task training process converge efficiently. Thus, we apply $l_2$ normalization to the output feature of FC layer.

### C. Domain Classifier

In an ideal feature space, the distribution of real speech is aggregated regardless of domain. Thus, a single-side adversarial domain classifier with Gradient Reverse Layer (GRL) [66] is proposed. As shown in Fig. 2, the domain discriminator

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TABLE I
LCNN BASED BACKBONE ARCHITECTURE

| Module | Filter/Stride | Output |
|---|---|---|
| Input | - | (batch,1,404,80) |
| Conv2d | 5 × 5 / 1 × 1 | (batch,64,404,80) |
| MFM | - | (batch,32,404,80) |
| MaxPool2d | 2 × 2 / 2 × 2 | (batch,32,202,40) |
| Conv2d | 1 × 1 / 1 × 1 | (batch,32,202,40) |
| MFM | - | (batch,32,202,40) |
| BatchNorm2d | - | (batch,32,202,40) |
| Conv2d | 3 × 3 / 1 × 1 | (batch,96,202,40) |
| MFM | - | (batch,48,202,40) |
| MaxPool2d | 2 × 2 / 2 × 2 | (batch,48,101,20) |
| BatchNorm2d | - | (batch,48,101,20) |
| Conv2d | 1 × 1 / 1 × 1 | (batch,96,101,20) |
| MFM | - | (batch,48,101,20) |
| BatchNorm2d | - | (batch,48,101,20) |
| Conv2d | 3 × 3 / 1 × 1 | (batch,128,101,20) |
| MFM | - | (batch,64,101,20) |
| MaxPool2d | - | (batch,64,50,10) |
| Conv2d | 1 × 1 / 1 × 1 | (batch,128,50,10) |
| MFM | - | (batch,64,50,10) |
| BatchNorm2d | - | (batch,64,50,10) |
| Conv2d | 3 × 3 / 1 × 1 | (batch,64,50,10) |
| MFM | - | (batch,32,50,10) |
| BatchNorm2d | - | (batch,32,50,10) |
| Conv2d | 1 × 1 / 1 × 1 | (batch,64,50,10) |
| MFM | - | (batch,32,50,10) |
| BatchNorm2d | - | (batch,32,50,10) |
| Conv2d | 3 × 3 / 1 × 1 | (batch,64,50,10) |
| MFM | - | (batch,64,50,10) |
| MaxPool2d | 2 × 2 / 2 × 2 | (batch,32,25,5) |
| Dropout | - | (batch,32,25,5) |
| BLSTM | - | (batch,25,160) |
| BLSTM | - | (batch,25,160) |
| FC & $l_2$ Norm | - | (batch,512) |
| FC & $l_2$ Norm | - | (batch,1) |

only discriminates the features from real domains. Let $p(X_r)$ denotes the distributions of real feature and $Y_D$ denotes the domain of $X_r$. The adversarial loss function of the domain discriminator is defined as follows:

$$\min_D \max_G L_{ada}(G, D) = -E_{x\sim P(X_r), y\sim Y_D} \sum_{d=1}^{3} p(y=d) log\left(D\left(G(x)\right)\right), \quad (1)$$

where $d$ denotes the domain label. Loss minimization in the discriminator and maximization in the generator occur simultaneously. The feature generator is trained to learn a robust feature to spoof the domain discriminator, thus maximizing the loss of $L_{ada}$ in the generator. In addition, the discriminator is trained to identify the feature domain by minimizing the loss of $L_{ada}$ in the discriminator. To achieve these goal simultaneously, the GRL is added. In the presence of the GRL, the discriminator is unable to identify which domain the real feature originates from. The GRL cannot be used in forwards propagation. Instead, the GRL reverses the gradient by multiplying negative dynamic coefficients in the backwards propagation, making the discrimination task difficult. As the number of training iterations iter increases, we set the negative dynamic coefficients $\lambda$, as introduced by [66], as follows:

$$\lambda = 1 - \frac{2}{1 + \exp(-k\frac{C_i}{T_i})}, \quad (2)$$

where $C_i$ means the current training iteration and $T_i$ denotes the total training iteration. In our experiment, we set the value of $k$ to 10. The negative coefficient decreases from 0 to -1 during the training procedure, so that the model will focus on optimizing the classifier loss at the beginning of the training process.

### D. Triplet Mining

Owing to the diversity of the spoofing method in ADD, the distribution of fake speech should be dispersed and far from the distribution of real speech in an ideal speech feature space, allowing straightforward discrimination and enabling categorization boundaries to be learned quickly. Moreover, the scattered speech features also facilitate the subsequent identification of a specific spoofing method. Therefore, the triplet mining [67] method is ideal for the aforementioned goals since it can aggregate the bonafide speech features while dispersing the spoofed features.

$$L_{tri} = \sum_{i}^{N} \left( \left\| f\left(x_i^a\right) - f\left(x_i^p\right) \right\|_2^2 - \left\| f\left(x_i^a\right) - f\left(x_i^n\right) \right\|_2^2 + \alpha \right), \quad (3)$$

where $x_i^a$, $x_i^p$, and $x_i^n$ represent the anchor sample, positive sample, and negative sample, respectively. The first term in Equation 3 computes the Euclidean distance between the anchor sample and positive sample, and the second term calculates the distance between the anchor sample and the negative sample. By minimizing $L_{tri}$, the distance between the anchor sample and the positive sample will decrease, and the anchor sample will move further away from the negative sample. We set $\alpha$ to 0.1, which is a margin value. In practice, we recombine the real and fake speech from the three different domains into four categories, as shown in the left corner of Fig. 2a. In our approach, all real speech samples within each batch are treated as collective positive samples, while the fake speech samples from the three domains are treated as separate negative samples. Through the above optimization process, we effectively disperse real and fake speech samples and further disperse the fake speech samples based on their respective domains.

### E. Loss Function

With the help of the single-side domain discriminator and triplet mining, the feature space of the training data is closer to the ideal feature space. In conclusion, the overall loss function $L_{all}$ is described in Equation 4.

$$L_{all} = L_{cls} + \lambda_1 L_{ada} + \lambda_2 L_{tri}, \quad (4)$$

where $L_{cls}$ utilizes the binary cross-entropy (BCE) to classify a feature as real or fake. Before using the BCE, an FC layer is added to reduce the dimension of the feature space from 512 to 1. It is worth noting that weight normalization is

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used in the FC layer; weight normalization is widely utilized in face recognition [68], [69] to maintain the stability of the gradient descent and improve the binary classification method.

## IV. Experiments

### A. Dataset

*1) Training and Validation Set:* We used three audio datasets, the "ASVspoof 2019 logical access (LA) database [70], WaveFake [21] and FakeAVCeleb [71]," as source domains for training. We chose these three well-known datasets because there are currently limited public datasets in ADD. ASVspoof2019 LA database is based upon the standard multi-speaker dataset named VCTK [72]. It was created utilizing 107 speakers (46 male, 61 female) utterances and contains a total number of 12,456 real and 108,978 fake utterances generated by using 11 TTS and 8 VC spoofing algorithms. We selected this dataset due to its diversity of spoofing types and popularity in ADD.

WaveFake is an audio deepfake dataset which created by 7 spoofing algorithms includding MelGAN [30], MelGAN Large, Parallel WaveGAN [31], Multi-band MelGAN [32], Full-band MelGAN, HiFi-GAN [33], and WaveGlow [73]. In our experiments, we maintained consistency in terms of the training dataset language by using only English during training, allowing us to assess the model performance across languages. Therefore, we opted not to include the JSUT dataset subset, which consists of Japanese language data. Instead, we used the LJSpeech subset as the training set, which consists of 13,100 real utterances and 91,700 fake utterances. The JSUT sub-dataset was used to evaluate the cross-language performance in our model.

FakeAVCeleb is a novel Audio-Video deepfake dataset that contains not only deepfake videos but also synthesized lip-synced fake audio. It includes 500 real and 11,357 fake videos by using one algorithm SV2TTS [74] which is a transfer learning-based real-time voice cloning tool. we extracted the audio track from the video in the data preprocessing step. We selected this dataset to discriminate not only audio but also audio tracks of videos in real scenarios.

The training and validation data include 26,056 bonafide utterances and 212,035 spoofing utterances. We used fold settings similar to those in [14], where different subsets of training data and validation data were used in each fold to increase the stability of the training and the generalizability of the feature model. Specifically, in all folds, we chose 80% of the real samples for training and 20% of the real samples for validation. For the fake samples, we selected different samples in each fold based on the types of attack methods, which are shown in Table II. For instance, in fold 1, the ASVspoof2019 validation set includes attack types A01-A04, while the remaining attack types A05-A19 are selected for training. It is worth mentioning that the FakeAVCeleb subset only has one attack type. To ensure smooth fold operations, we used five folds based on the types of video manipulation techniques. We guarantee that for each validation, data are unseen in the training process to achieve good model generalizability to the target domain.

TABLE II
Fold Setting in Validation Dataset

| Fold | ASVSpoof2019 | WaveFake | FakeAVceleb |
|---|---|---|---|
| 1 | A01, A02, A03, A04 | MelGAN-large | FSGAN-wav2lip |
| 2 | A05, A06, A07, A08 | Parallel WaveGAN | RTVC |
| 3 | A09, A10, A11, A12 | Multi-band MelGAN | FaceSwap |
| 4 | A13, A14, A15, A16 | MelGAN, WaveGlow | Wav2lip |
| 5 | A17, A18, A19 | HifiGAN, FB-MelGAN | Wav2lip |

*2) Evaluation Set:* To verify the classification performance of our model on unknown target domains, we test our model in harsh conditions: cross-language and noisy datasets which include FAD [75] and JSUT [76]. FAD is a Chinese dataset for fake audio detection. Six real speech sources were selected from FAD, R1: AISHELL1 [77], R2: AISHELL3 [78], R3: THCHS-30 [79], and R4: MAGICDATA Mandarin Chinese Read Speech Corpus. R5: MAGICDATA Mandarin Chinese Conversational Speech Corpus [80] and R6: Self-Recording Corpus. Twelve mainstream speech generation techniques were used to generate fake audio. Furthermore, three noise datasets, PNL 100 Nonspeech Sounds [81], NOISEX92 [82] and TAU Urban Acoustic Scenes [83], were chosen for the addition of noise at five different signal-to-noise ratios (SNRs), 0 dB, 5 dB, 10 dB, 15 dB, and 20 dB, to simulate real-world scenarios.

For our evaluation dataset, we chose eight different attack types in FAD, C1: LPCNet [84], C2: StyleGAN [85], C3: MB-MelGAN [32], C4: Parallel WaveGAN [31], C5: Hifi-GAN [33], C6: WaveNet [23], C7: Tacotron-HifiGAN [24], and C8: FastSpeech-HifiGAN [27]. To test the classification ability of the model in a real scenario, N1-N8 were chosen. N1-N8 are the C1-C8 test sets with added noise. Specifically, N1-N6 were obtained by adding noise from the PNL 100 Nonspeech Sounds dataset to C1-C6, while N7 and N8 were obtained by adding noise from the NOISEX92 and TAU Urban Acoustic Scenes datasets to C7 and C8, respectively. For C1-C8 and N1-N8, we randomly chose 3,000 real and 3,000 fake utterances in each condition. For the 3000 real utterances, we selected 500 utterances in each real source to increase the diversity in the source domain. C9 is a JSUT subdataset that includes 1,000 bonafide utterances and 2,000 spoof utterances (1,000 utterances from MB-MelGAN and 1,000 from Parallel WaveGAN). The details of the dataset construction can be seen in Fig. 3.

### B. Implementation Details

To maintain consistency among the input data across different test methods, all audio files were resampled to 16 kHz and trimmed or zero-padded to 4s. It should be noted that in our experiment, we deliberately abstained from employing any DA techniques to introduce additional data, relying solely on training with the data from the three source domains.

For the baselines RawNet2, RawGAT-ST and AASIST, the input is the raw waveform of 4s (64,000 samples). The input of the other baseline methods in Table III is 80 dimensions of LFCC. All methods utilize 5 folds for training, and the fold

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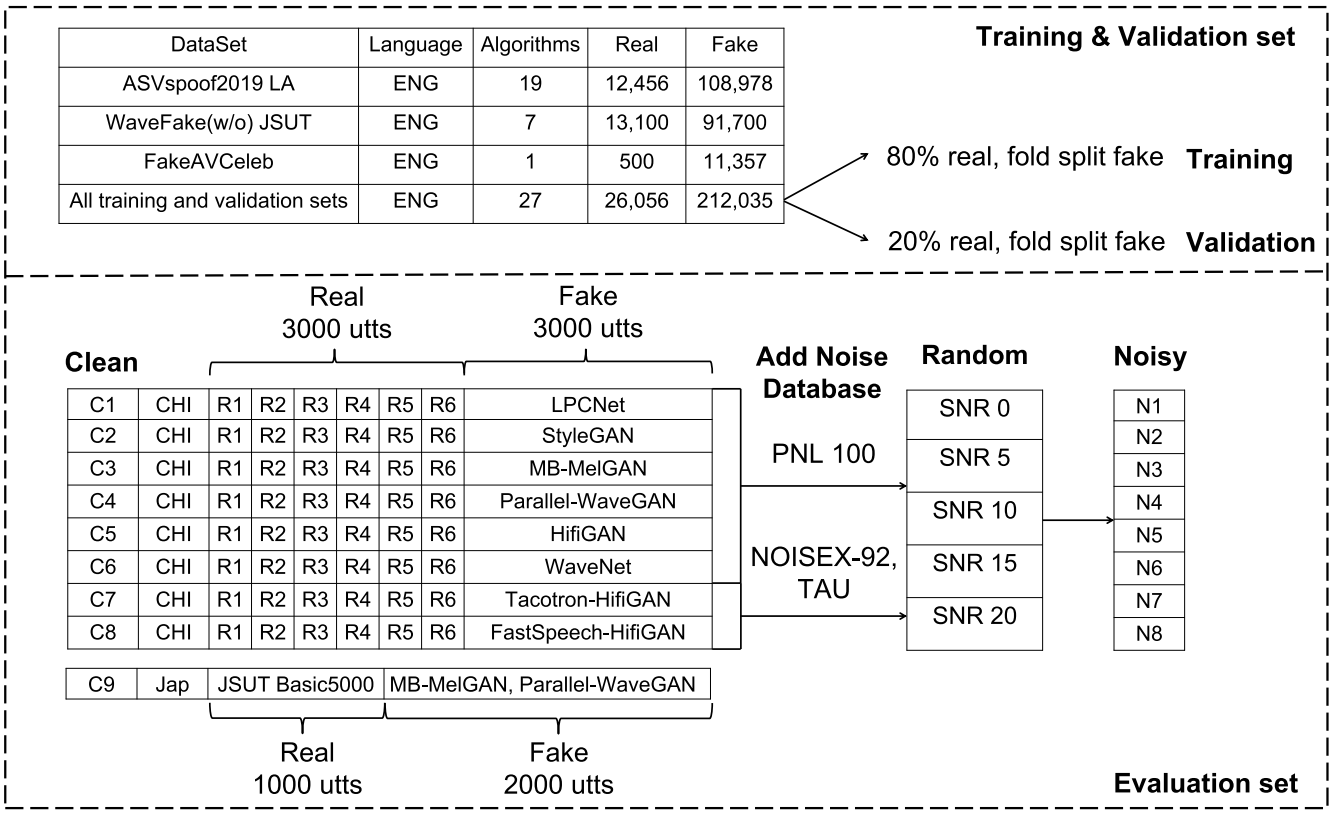


Fig. 3. Datasets Partition and Construction. Top demonstrates the training and validation datasets, while the bottom part displays the evaluation set used in our experiments, which includes conditions C1-C9 and noise conditions N1-N8.

TABLE III
EER EVALUATION FOR C1-C9 TEST SETS

| Model | Features | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | AVG |
|---|---|---|---|---|---|---|---|---|---|---|---|
| GMM [70] | LFCC | 18.26 | 13.20 | 15.67 | 12.78 | 32.57 | 27.23 | 18.69 | 25.29 | 32.92 | 21.85 |
| XceptionNet [86] | LFCC | 13.36 | 13.08 | 6.23 | 12.83 | 16.80 | 24.60 | 5.06 | 6.86 | 26.43 | 13.92 |
| MesoInception [87] | LFCC | 20.03 | 14.43 | 7.90 | 5.73 | 34.90 | 32.50 | 5.56 | 24.95 | 21.20 | 18.58 |
| Resnet18 [10] | LFCC | 11.60 | 9.94 | 6.63 | 2.71 | 18.67 | 24.33 | 5.20 | 5.89 | 23.80 | 12.09 |
| RawNet2 [6] | Audio | 33.23 | 32.92 | 39.50 | 38.88 | 64.10 | 42.60 | 40.20 | 65.43 | 47.60 | 44.94 |
| RawGAT-ST [8] | Audio | 21.84 | 21.41 | 25.32 | 19.25 | 35.73 | 31.22 | 28.23 | 37.48 | 34.98 | 28.38 |
| AASIST [56] | Audio | 11.67 | 15.23 | 28.23 | 20.00 | 33.70 | 15.70 | 28.96 | 22.88 | 6.98 | 20.37 |
| ASDG-R | LFCC | 3.30 | 12.93 | 10.26 | 4.33 | 7.00 | 10.60 | 9.06 | 11.06 | 4.10 | 8.07 |
| ASDG-E | LFCC | 2.67 | 10.18 | **3.60** | **0.70** | 8.13 | 11.78 | 4.36 | 10.56 | **3.94** | 6.21 |
| ASDG | LFCC | **2.43** | **8.16** | 4.25 | 1.30 | **6.26** | **8.56** | **2.76** | **5.44** | 12.14 | **5.70** |

settings are the same. In each fold, we trained for 5 epochs with validation. For the training strategy, the Adam optimizer was adopted, and $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-9}$ and the weight decay was set to $10^{-4}$. The batch size was set to 96 in each model. Specifically, our data loader retrieves training data from six categories, including domain 1 real and fake, domain 2 real and fake, and domain 3 real and fake. Each category consists of 16 audio samples; thus, there are a total of 96 audio samples per batch. The hyperparameter of the loss function for $\lambda_1$ and $\lambda_2$ was set to 0.1 to balance the proportion of different loss functions.

### C. Cross-Dataset Experiments in Clean Conditions

*1) Compare to The Baseline Models:* In this section, we tested our model with the evaluation datasets, as mentioned in Section III-A. The training set comprises three English domain datasets, while the testing set includes diverse domains such as Chinese (C1-C8) and Japanese (C9). Therefore, our experiments encompass cross-domain, cross-language, and unseen attack scenarios, effectively validating the model generalizability. The equal error rate (EER) of each baseline and our model is shown in Table III. ASDG represents our proposed method, which achieves the best performance, with a 5.70% average EER. Although our training dataset is rich and contains three source domains, the average EER of the other baseline models for the unknown target is still more than 12%. To further evaluate the performance of our ASDG and other baseline models in out-of-domain scenarios, we conducted additional tests using various metrics on ResNet18, RawNet2, AASIST, and ASDG.

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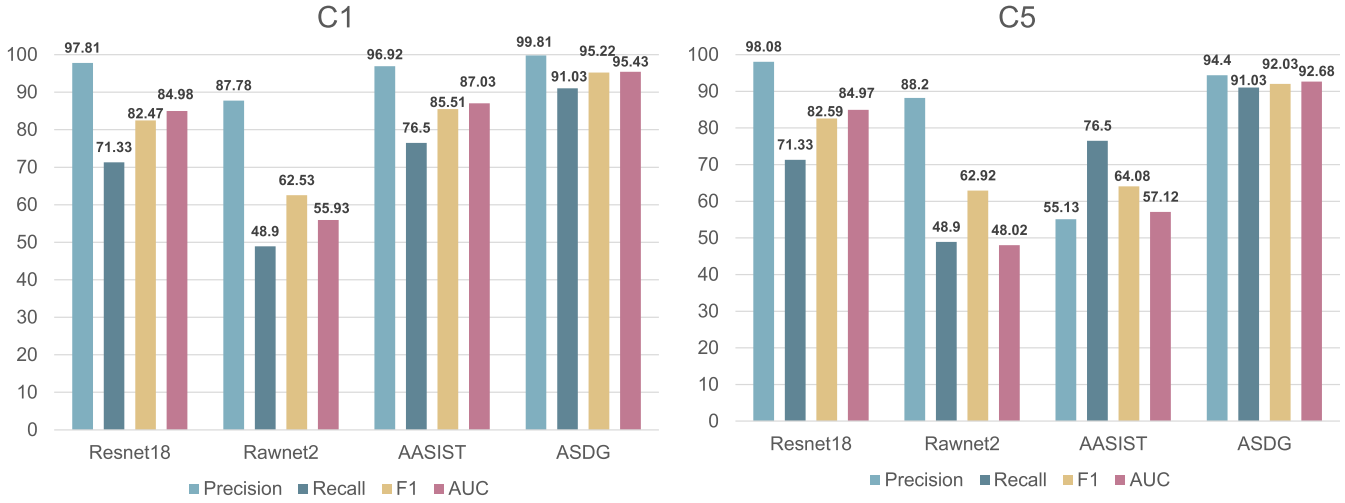


Fig. 4. Four evaluation metrics for evaluate the model performance with C1 and C5.

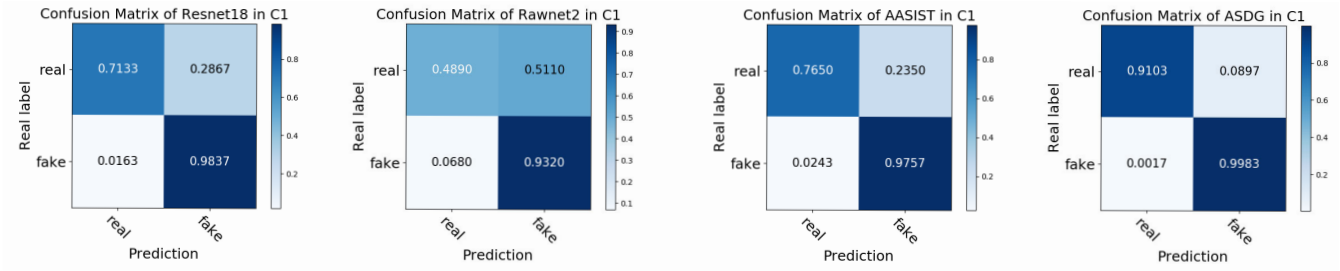


Fig. 5. The confusion matrices of baseline methods and our ASDG for the C1 testset.

In Fig. 4, we tested four different metrics other than EER for C1 and C5: precision, recall, F1-score and area under the curve (AUC). Specifically, precision and recall indicate the fraction of relevant instances among the retrieved instances and the fraction of real data recognized as correct, respectively. The F1-score combines precision and recall into a single value, providing a balanced assessment of the model performance. The AUC represents the area under the receiver operating characteristic (ROC) curve, which is a plot of the true positive rate against the false-positive rate at various classification thresholds. In our experiments, we set the above evaluation metrics' threshold to 0.5. By considering the above four metrics, we can gain a comprehensive understanding of the model's effectiveness in the ADD task.

From the evaluation results, we noted that the model with poor EER metrics also usually has low recall metrics. For instance, the highest EER in Table III is obtained using the RawNet2 model, which achieves recall values of 48.9% for both C1 and C5. For the ResNet18 model, exceptional precision is achieved, but the low recall of 71.33% diminishes its overall performance. As a result, ResNet18 achieves an EER of 11.60% and an ROC of 84.98%. For the AASIST model, the recall remains relatively low at 76.5%, and for C5, the precision of 55.13% is significantly low, indicating that the model tends to incorrectly classify a substantial number of genuine speech samples as fake, particularly when utilizing raw audio as input. However, in our model, the recall values for C1 and C5 are both 91.35%, a significant increase compared to that of the model without the DG strategy. To further investigate our improvements, we present the confusion matrices of our ASDG and the baselines in Fig. 5. From the confusion matrices, we further observed that ResNet18, RawNet2, and AASIST have high false-positive (FP) values in the top-right corner. Specifically, this indicates that these models misclassify 28.67%, 51.10%, and 23.50% of originally genuine speech as fake speech, respectively. On the other hand, our ASDG misclassifies only 8.97% of genuine speech as fake speech, demonstrating the excellent discriminative ability of our model towards genuine speech. In real-world detection tasks, the occurrence of genuine speech tends to outnumber that of fake speech, making high recall particularly crucial.

To test the effect of the fold setting on the out-of-domain ADD task, we selected the best checkpoints for each fold (folds 1-5) and used these checkpoints to determine the EERs for C1, as shown in Fig. 6. The results demonstrate that with an increase in the training epochs of each fold, the antispoofing performance in unknown domains continues to improve.

*2) Compare to Different Backbone:* To examine whether the LCNN-BLSTM is suitable for the ASDG strategy, we attempted to employ other backbones for testing. We selected ResNet18 [55] and ECAPA-TDNN [88] as alternative backbones. As observed at the bottom of Table III, ASDG-R and ASDG-E are the ResNet18 and ECAPA-TDNN

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TABLE IV
EER RESULTS IN NOISY CONDITIONS

| Model | Feature | N1 | N2 | N3 | N4 | N5 | N6 | N7 | N8 | AVG |
|---|---|---|---|---|---|---|---|---|---|---|
| GMM [70] | LFCC | 38.80 | 32.67 | 37.56 | 42.28 | 47.36 | 37.87 | 38.80 | 32.32 | 38.46 |
| XceptionNet [86] | LFCC | 35.56 | 27.73 | **18.66** | **23.48** | 18.56 | 36.90 | 27.08 | 20.10 | 26.01 |
| MesoInception [87] | LFCC | 40.86 | 31.36 | 31.24 | 29.70 | 43.50 | 43.80 | 12.53 | 28.43 | 32.68 |
| Resnet18 [55] | LFCC | 36.36 | 24.95 | 25.63 | 26.50 | 21.53 | 32.53 | 22.40 | 23.36 | 26.66 |
| RawNet2 [6] | Audio | 40.63 | 38.96 | 38.63 | 43.78 | 63.68 | 53.13 | 40.80 | 54.86 | 46.81 |
| RawGAT-ST [8] | Audio | 39.67 | 32.65 | 34.80 | 38.60 | 42.21 | 43.67 | 37.58 | 52.77 | 40.24 |
| AASIST [56] | Audio | **27.13** | 25.86 | 29.63 | 31.26 | 38.71 | 31.43 | 36.33 | 50.38 | 33.84 |
| ASDG-R | LFCC | 32.01 | 29.91 | 34.13 | 28.80 | **18.51** | **26.30** | **9.13** | **11.70** | **23.81** |
| ASDG-E | LFCC | 33.67 | 28.35 | 30.34 | 31.80 | 38.08 | 39.28 | 36.95 | 31.90 | 30.04 |
| ASDG | LFCC | 35.25 | **24.93** | 32.91 | 31.07 | 31.92 | 36.60 | 18.12 | 25.01 | 29.48 |

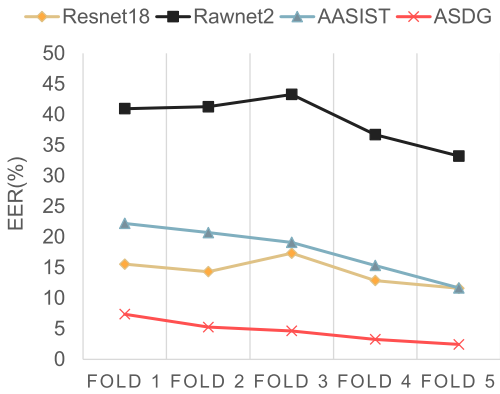


Fig. 6. The EER(%) results in different folds of the baseline methods and our ASDG for the C1 testset.

TABLE V
THE NUMBER OF TRAINABLE PARAMETERS IN EACH EVALUATED ARCHITECTURE

| Backbone | Trainable parameters |
|---|---|
| LCNN-BLSTM | 549,696 |
| ResNet18 | 12,974,320 |
| ECAPA-TDNN | 6,388,675 |

backbones with the ASDG strategy, respectively. The EERs decrease by 2.37% and 0.51% using the ASDG model compared to those obtained using the ASDG-R and ASDG-E models with clean conditions, respectively. ASDG-E shows promising results and surpasses ASDG for C3, C4 and C9. However, since applying the DG method requires a consideration of the efficiency (parameters) of the backbone network, it is important to evaluate the parameter count of different backbone networks. In Table V, we compare the training parameters of three backbone networks. It is evident that when the LCNN is used as the backbone, the number of trainable parameters is only 4.24% and 8.60% of those used in ResNet18 and ECAPA-TDNN, respectively. The lightweight design of the LCNN structure enables the easy integration of various DG methods, which is crucial for conducting extensive research on scalability and advancing its capabilities in out-of-domain ADD tasks, as demonstrated by its good EER performance.

### D. Harsh Condition

In Table IV, we tested our model in noisy conditions. N1-N8 are the harshest conditions, and they include unseen attacks as well as cross-language and random noisy settings. ASDG-R achieves the best performance in these conditions, while the ASDG model ranks fourth among the baseline models. The test results in harsh conditions are not stable without any noise augmentation methods. To further analyse the performance of the models in harsh conditions, we plotted the confusion matrix graphs for the abovementioned four models for N2, as shown in Fig. 7. For N2, although ASDG achieves the lowest EER of 24.93%, it only outperforms the second best model, ResNet18, by a marginal improvement of 0.02%. In the confusion matrix, we observed that the inferior performance of the ASDG model is attributed to a high false-negative (FN) rate (lower left corner), indicating the misclassification of fake speech as genuine. However, we emphasize that in the context of practical antispoofing tasks, where genuine speech is predominant, the false-positive (FP) value holds greater importance. It is worth noting that the genuine speech samples (N1-N8) remain unchanged across the confusion matrices, resulting in consistent true-positive (TP) and FP values (the first row of the confusion matrix). As depicted in Fig. 7, ASDG exhibits the lowest FP value, misclassifying only 10.27% of genuine speech as fake, demonstrating the model's superior ability to accurately classify genuine speech compared to that of the baseline models.

### E. Loss Function Study

*1) Ablation Study of Loss Functions:* To verify the importance of triplet mining and domain adversarial learning, an ablation study was conducted, and the results are shown in Table VI. ASDG(cls) adopts a feature generator using a classifier and the BCE to identify speech. ASDG(cls+tri) and ASDG(cls+ada) adopt triplet mining alone and domain adversarial learning alone, respectively. Compared to the EER achieved using the ASDG(cls) approach, the EER decreases by 7.70% with the triplet mining method and 8.14% with the domain adversarial learning approach. ASDG(cls+ada+tri) is our proposed model, which has the lowest EER. When using this model, the EER is reduced by 11.11% compared to the EER obtained using ASDG(cls).

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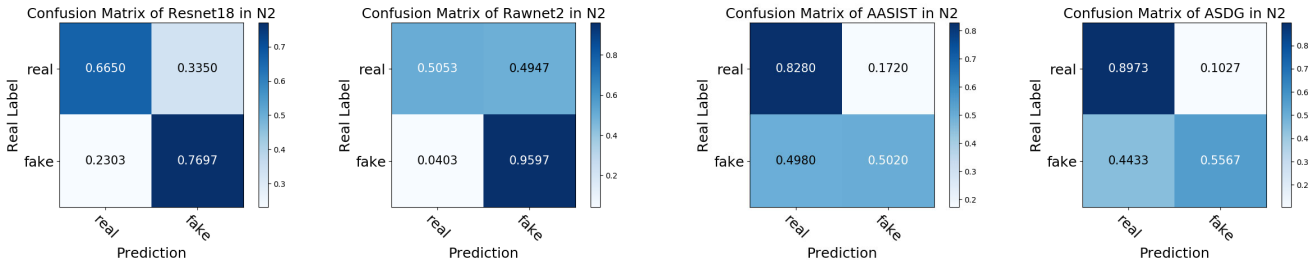


Fig. 7. The confusion matrices of the baseline methods and our ASDG for the N2 testset.

TABLE VI
THE PERFORMANCE OF DIFFERENT LOSS FUNCTIONS FOR ASDG

| Model | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | AVG |
|---|---|---|---|---|---|---|---|---|---|---|
| ASDG(cls) | 11.80 | 24.25 | 19.33 | 8.44 | 13.36 | 19.67 | 14.42 | 13.80 | 24.22 | 16.81 |
| ASDG(OC-Softmax) | 31.13 | 9.86 | **2.32** | 1.94 | 7.70 | **5.56** | 2.85 | **2.65** | 21.88 | 9.54 |
| ASDG(cls+tri) | 6.16 | 10.72 | 2.86 | 2.46 | 7.41 | 17.49 | 4.56 | 9.13 | 21.17 | 9.11 |
| ASDG(cls+ada) | 3.90 | 13.25 | 6.46 | 2.32 | 8.97 | 16.50 | 3.10 | 4.30 | 19.22 | 8.67 |
| ASDG(cls+ada+tri) | **2.43** | **8.16** | 4.25 | **1.30** | **6.26** | 8.56 | **2.76** | 5.44 | **12.14** | **5.70** |

TABLE VII
EER RESULTS OF ASDG WITH DIFFERENT INPUT FEATURES

| Feature | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | AVG |
|---|---|---|---|---|---|---|---|---|---|---|
| MelSpec | 21.20 | 26.36 | 16.06 | 11.93 | 18.89 | 18.37 | 18.83 | 31.23 | 50.00 | 23.65 |
| MFCC | 52.60 | 44.20 | 53.18 | 41.73 | 34.04 | 42.72 | 53.20 | 10.20 | 33.98 | 40.65 |
| IMFCC | 20.60 | 12.36 | 13.56 | 4.36 | 27.10 | 21.10 | 6.30 | 26.60 | **3.30** | 15.03 |
| CQCC | 29.83 | 49.90 | 41.06 | 40.53 | 36.24 | 34.40 | 49.43 | 37.63 | 36.20 | 39.47 |
| LFCC | **2.43** | **8.16** | 4.25 | **1.30** | **6.26** | 8.56 | 2.76 | 5.44 | 12.14 | **5.70** |
| MFCC+MelSpec | 49.56 | 50.05 | 50.08 | 49.08 | 50.08 | 48.85 | 49.57 | 50.12 | 50.08 | 49.72 |
| LFCC+MelSpec | 10.04 | 15.98 | **2.53** | 1.63 | 9.70 | 11.56 | **1.86** | 11.82 | 19.10 | 9.36 |
| LFCC+MFCC | 22.21 | 8.21 | 19.87 | 18.75 | 20.13 | 39.26 | 16.53 | 8.70 | 21.30 | 19.44 |
| W2V2-XLS-R | 6.93 | 6.16 | 6.67 | 6.16 | 14.67 | **5.67** | 7.00 | 14.76 | 5.20 | 8.14 |

*2) Compare to Different Loss Functions:* Our ASDG improves the generalizability by aggregating the distribution of real speech regardless of domain. We note that our idea is similar to the OC-Softmax method [55], which treats the ADD task as a one-class learning problem to enhance the model's generalizability. Specifically, a loss function named OC-Softmax was proposed to train a feature space where genuine speech embeddings form a compact boundary, while spoofing data are separated from genuine data by a certain margin. We utilized the same settings as those in the original paper and replaced the final BCE loss function with the OC-Softmax. As shown in Table VI, ASDG(OC-Softmax) is a one-class learning method that shows certain advantages and outperforms ASDG for C3, C6, and C8. However, its performance is less favorable for C1 and C9, affecting its overall performance.

### *F. Feature Study*

For a forensic system to have good generalizability, both a generalized model design and appropriate features are indispensable. In this section, we investigated the impact of different features on the model performance, including traditional handcrafted features, fused handcrafted features and self-supervised pretrained features, as shown in Table VII.

*1) Compare to Single Handcrafted Feature:* For single handcrafted feature, we utilized 80-dimensional Mel spectrogram (Melspec), MFCC, IMFCC [89], LFCC and 20-dimensional CQCC [90] features in our study.

The worst performance is obtained using MFCC features in cross-dataset conditions. We further observed that with the MFCC features, almost all speech is identified as bonafide, which proves that the MFCC features are not generalized and thus not appropriate in our out-of-domain conditions. A mel-spectrogram is used for the target features in most audio spoofing attack models. Both the mel-spectrogram and MFCC are generated from a Mel-scaled filter, which focuses on obtaining high-level low-frequency information to deceive the sense of hearing. Meanwhile, these spoofing models may show inadequacy in the presence of high-frequency information, which may lead to the overfitting of the low-frequency feature to the training set, resulting in a loss of generalization.

Unlike MFCC features, IMFCC features utilize an inverted mel-scaled filter bank and focus on high-frequency features. Therefore, we employ IMFCC features to test the antispoofing capability. The results indicate that IMFCC features rank second among the handcrafted features, particularly leading to the lowest EER for C9. However, there is still a slight gap in performance compared to that with the LFCC features

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TABLE VIII
EER RESULTS OF DIFFERENT DIMENSION LFCC

| Feature | dim | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | AVG |
|---|---|---|---|---|---|---|---|---|---|---|---|
| LFCC | 20 | 29.55 | 40.03 | 56.23 | 35.45 | 12.70 | 28.86 | 56.32 | 9.43 | 9.10 | 30.85 |
| LFCC | 40 | 33.18 | 21.33 | 17.41 | 18.68 | 41.16 | 42.96 | 17.42 | 34.52 | 27.74 | 28.27 |
| LFCC | 60 | 10.00 | 8.46 | 4.41 | **1.30** | 10.38 | 12.70 | **1.83** | 9.30 | 16.60 | 8.33 |
| LFCC | 80 | **2.43** | **8.16** | **4.25** | **1.30** | **6.26** | **8.56** | 2.76 | **5.44** | 12.14 | **5.70** |
| LFCC | 100 | 6.36 | 21.65 | 7.92 | 5.76 | 10.10 | 14.75 | 4.46 | 10.10 | **6.60** | 9.74 |

for C1-C8. Through analysis, we found that with the IMFCC features, there is a tendency to misclassify many genuine utterances as fake. While the IMFCC features capture the differences between genuine and fake speech by focusing on high-frequency information, neglecting low-frequency information may affect the ability to discriminate out-of-domain genuine speech.

The results show that with the CQCC features, the performance is poor, with only a 1.18% improvement over that of the MFCC features. Similar to the MFCC features, the CQCC features utilize the constant-Q transform (CQT) based on the human auditory system's characteristics, with higher resolution in the low-frequency range. Thus, the high-frequency range is neglected. This may lead to overfitting of the model to the training domain and a decrease in the out-of-domain performance.

The performance of the model with the single LFCC feature is the best, with an EER of 5.70%. The LFCC feature maintains a balance of high and low frequency information, which is a good choice for out-of-domain ADD.

*2) Fusion of Handcrafted Features:* Considering the divergent attentional orientations of various manual features, we hypothesized that the fusion of handcrafted features may enhance the performance of ADD. The results indicate that the models using the three different fusion approaches did not outperform the model utilizing LFCC alone. It should be noted that LFCC+MelSpec achieved the lowest EER for C3 and C7, surpassing the performance of LFCC alone. This demonstrates that feature concatenation can complement the performance of a single feature in certain situations.

*3) Compare to Self-Supervised Features:* Many studies have demonstrated the excellent performance of self-supervised models as front ends in ADD. In our feature study, we utilized Wav2Vec2-XLS-R [91] as the front end for testing. Specifically, we extracted the 1024-dimensional last hidden states of the pretrained W2V2 by the Hugging face version of Wav2Vec2-XLS-R-300M[1] and froze the weights of the front end. The front-end model was pretrained with 436k hours of unannotated genuine speech data in 128 languages. The result can be found in the last row of Table VII. With W2V2 features, the model performs moderately well for all conditions and exhibits the second lowest EER of 8.14%. The generalizability of W2V2 may require further validation with more complex unseen domains. However, this method employs features that have been pretrained with large datasets, thereby incorporating additional source domain data. As the primary objective of this study was to evaluate the effectiveness of the ASDG method within a constrained domain, we will refrain from an extensive discussion regarding self-supervised features.

*4) Analysis of LFCC Features at Different Dimensions:* To further analyse why the use of a single LFCC leads to such promising performance under cross-domain conditions, we first investigated the impact of LFCC dimensions on EER, as shown in Table VIII. In previous studies on ADD, many researchers have explored the influence of different features within the same domain [92]. However, two aspects have been overlooked. First, the performance of different features may vary under cross-domain conditions. Second, the specific parameters of each feature have not been thoroughly investigated. For coefficient-based features such as MFCC, the final step involves applying a discrete cosine transform (DCT) and selecting a corresponding number of dimensions. For example, in speech recognition tasks, MFCC often use 13 dimensions. However, in the case of ADD, especially cross-domain scenarios, more dimensions, specifically higher-frequency information, are often required for joint decision-making.

In Table VIII, when 20-dimensional and 40-dimensional LFCC features are used, the performance is not satisfactory, with EERs of 30.85% and 28.27%, respectively. However, when the LFCC dimension was increased to 60, the features capture more high-frequency information that is sufficient for distinguishing between genuine and fake speech, resulting in a decreased EER of 8.33%. In our experimental conditions, the model achieved the best EER of 5.70% when the LFCC dimension was set to 80. When the dimension was increased to 100, a performance drop was observed for C1 to C8. However, the best EER was obtained for C9. We hypothesize that the performance drop for features exceeding 80 dimensions may be due to the need for a more powerful backbone network to capture fine-grained details and the requirement for more training epochs.

To further investigate why single LFCC features lead to good performance in the target domain, we visualized the LFCC features for genuine and spoofed speech. We selected the C5 condition, which only utilizes one of the most popular vocoders, HiFi-GAN, to regenerate genuine speech. With the synthesized speech generated by HiFi-GAN, a mean opinion score (MOS) of 4.36 was achieved, which is slightly below the MOS of 4.45 for real speech. This outcome indicates that fake speech is perceived by human listeners as nearly indistinguishable from real speech. However, differences still exist between genuine and spoofed speech in high-dimensional LFCC features, particularly at silent frame positions.

[1] https://huggingface.co/facebook/wav2vec2-xls-r-300m

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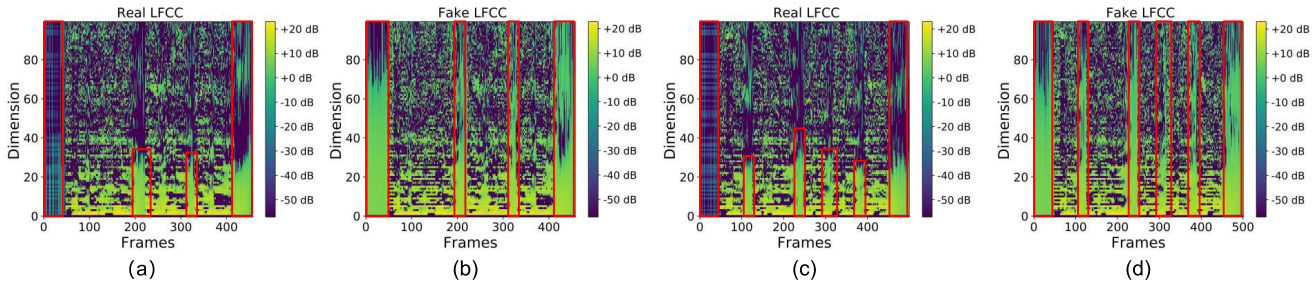


Fig. 8. Visualization of 100 dimensional LFCC of real and fake speech for C5. (a) and (c) represent genuine speech, while (b) and (d) represent spoofed speech. The red rectangular boxes highlight the fundamental differences between genuine and spoofed speech, where is the silent frame position.

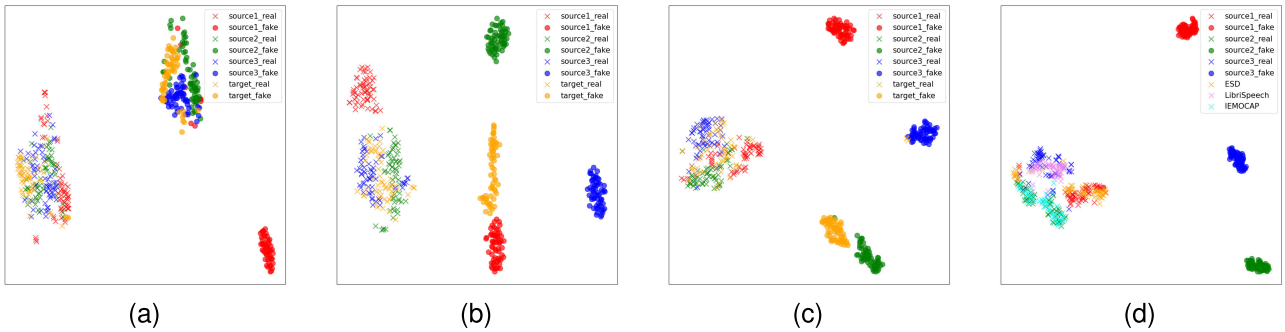


Fig. 9. Visualization of the feature space. The graphs of (a), (b), and (c) show the T-SNE visualization of ASDG (cls), ASDG (cls+tri), ASDG (cls+tri+ada) respectively. Different colors indicate features from different domains: red=ASVSpoof 2019LA, green = WaveFake and blue = FakeAVCeleb. Different shapes represent different categories information: point=fake and cross=real.

In Fig. 8, (a) and (c) represent genuine speech and exhibit distinct silence positions due to different speech pause points. In the beginning and ending positions of speech, there are usually no "energy columns" or only low-energy vertically discontinuous energy columns. At the silence position within the two real utterances, the peak of the energy column usually does not exceed 40 dimensions. However, in the spoofed speech samples (b) and (d), the energy columns at the beginning and end of the utterance significantly increase in both intensity and height. The peak of the energy columns at the beginning of the sentence reaches approximately 80 dimensions.

Moreover, at pause points within the sentence, the peak of the energy column greatly increases, forming a continuous vertical pattern. In nonsilence positions, there are still differences between genuine and spoofed speech, such as the density of LFCC energy points and their duration. However, these differences are relatively small compared to the differences at silent frames. We suppose that LFCC utilize these universal differences at silent frames for out-of-domain detection. Furthermore, since real speech also exhibits energy columns below 40 dimensions, low-dimensional LFCC does not have good discriminative power. As the energy height of the initial silent frames is often within 80 dimensions, an 80-dimensional LFCC is a suitable choice.

### *G. Visualization of The Feature Space*

To analyse the feature space learned by our model and verify the effectiveness of the triplet mining and adversarial learning, we visualize the distribution of different features using T-SNE [93]. As shown in Fig. 9, we randomly selected 420 samples from four data domains that represent source domains 1-3 and the target domain. In each domain, we selected 60 samples for real utterances and 60 for fake utterances. The target domain data are the C1 data, which are unseen for the training and validation. By adding the triplet loss, the distribution of real speech in the source domain starts to converge gradually, and the fake speech begins to disperse. After combining the triplet loss and adversarial loss, the distribution of real speech is more aggregated and entangled than that of adding triplets alone. Moreover, the real speech from the target domain is mapped to an entangled real speech cluster, and the fake speech forms a separate cluster as a result of the new attack type. The final feature space conforms with our view of the ideal feature space in the ADD task, which enhances the interpretability of hidden states.

Due to the limited availability of ADD datasets, we turned to collections of real speech datasets, namely, LibriSpeech [94], IEMOCAP [95], and ESD [96]. In the case of LibriSpeech, we randomly selected six speakers and chose ten utterances from each speaker for testing purposes. For the emotion datasets, IEMOCAP and ESD, we specifically selected five emotions, neutral, happy, angry, sad, and surprise, and selected twelve utterances for each emotion. These selections were made to ensure a diverse representation of real speech data across various speakers and emotional states. As shown in Fig. 9d, the 180 real speech samples from the three tested domains, along with the three real domains in the source domain, formed a large cluster. This observation indicates that our model is capable of effectively clustering out-of-domain real speech. The formation of such a cluster further validates

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the ability of our model to generalize well to real speech data from different domains.

## V. Conclusion

In this paper, we introduce an aggregation and separation domain generalization method and apply it to the ADD task. We use a modified LCNN as the backbone to distinguish real and fake images. In addition, a triplet mining method is designed to separate fake speech and aggregate real speech. Meanwhile, the single-side domain classifier makes the real speech from different domains undistinguishable for the further aggregation of real speech.

The difference between the current ADD model and ours is that we focus on improving the model generalizability, that is, achieving good performance even in unknown domains. We expect to generate an ideal ADD feature space in a limited dataset and achieve that goal by visualizing the feature space. According to the test results, our proposed model demonstrates the best performance in the out-of-domain ADD task, achieving the lowest EER of 5.70% in clean conditions. Future work will concentrate on more applications of DG and transfer learning methods for ADD task.

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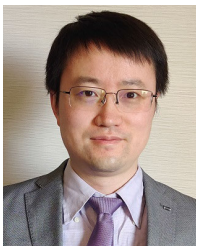

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